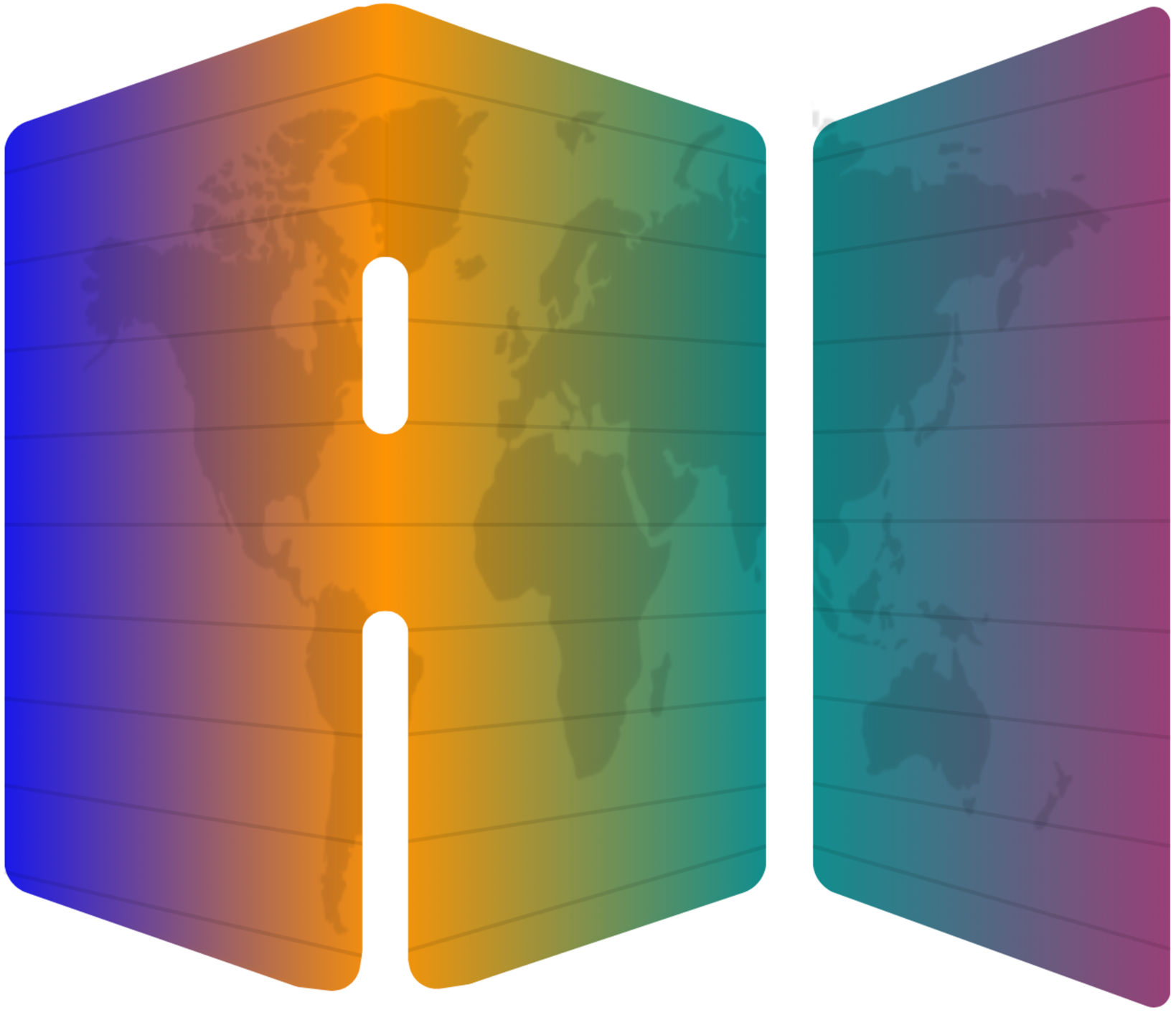
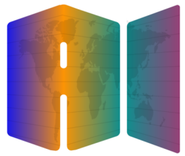


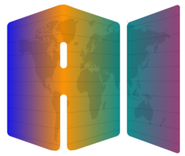
Model Context Protocol

Lectures by Elie Schoppik



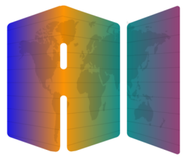
Model Context Protocol

Why MCP



Why MCP

Models are only as good as the
context given to them



Why MCP

What is the Model Context Protocol (MCP)

MCP is an open protocol that standardizes how your **LLM applications** connect to and work with your **tools & data sources.**

REST APIs

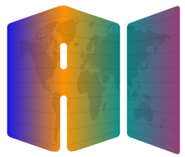
Standardize how **web applications** interact with the **backend**

LSP

Standardizes how **IDEs** interact with **language-specific tools**

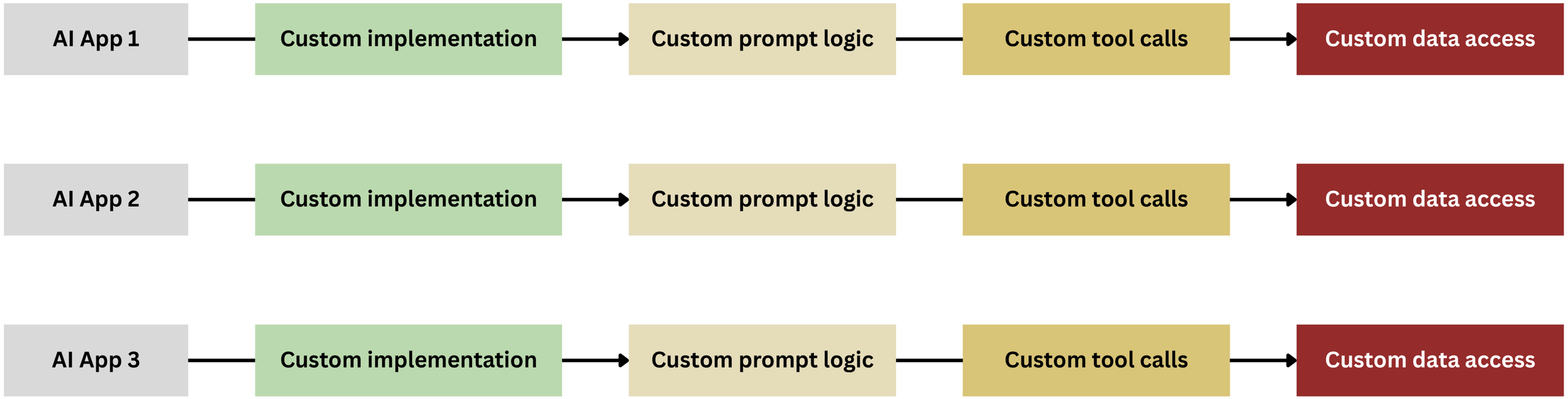
MCP

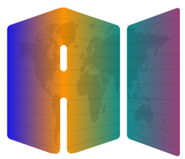
Standardizes how **AI applications** interact with **external systems**



Why MCP

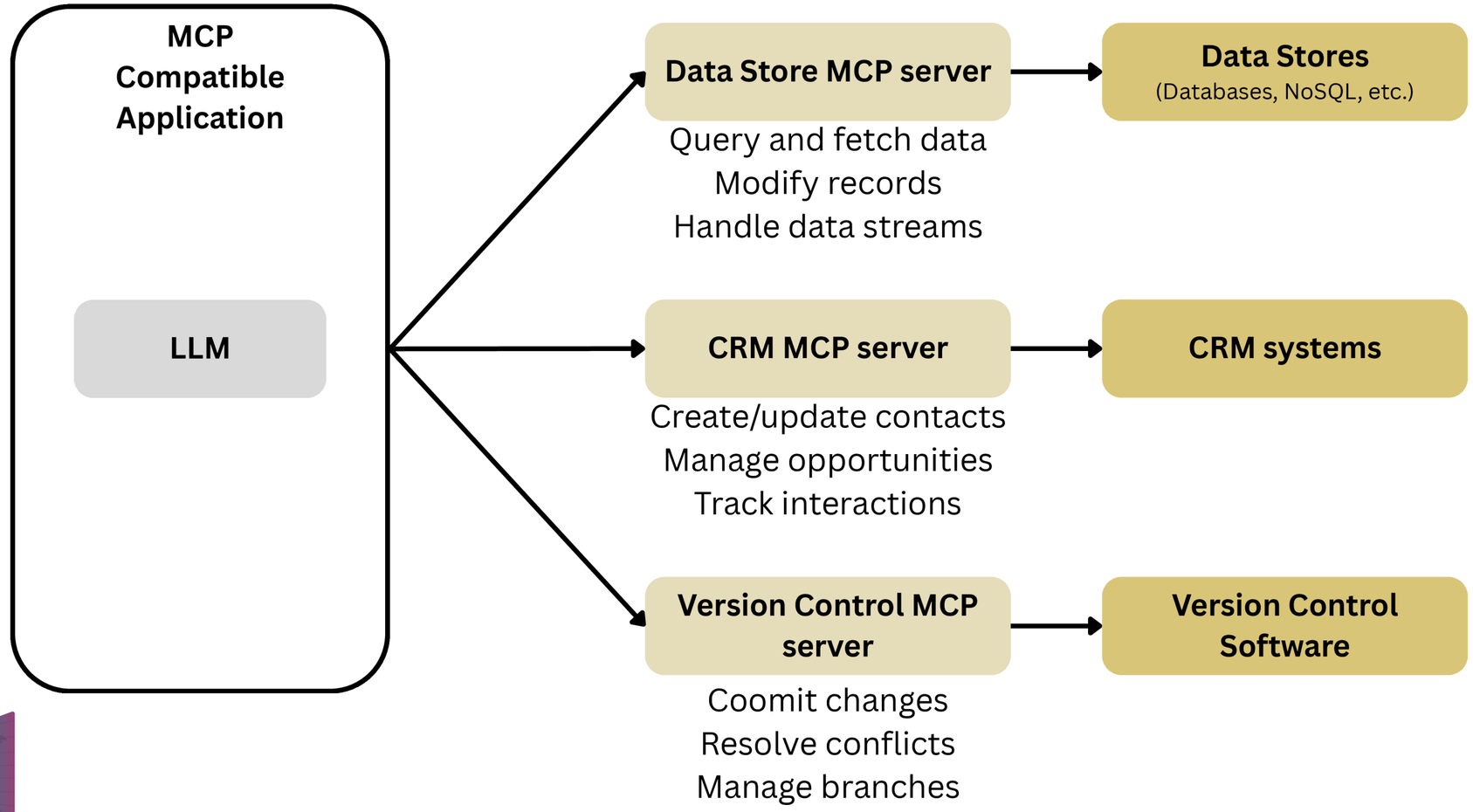
Without MCP: Fragmented AI Development





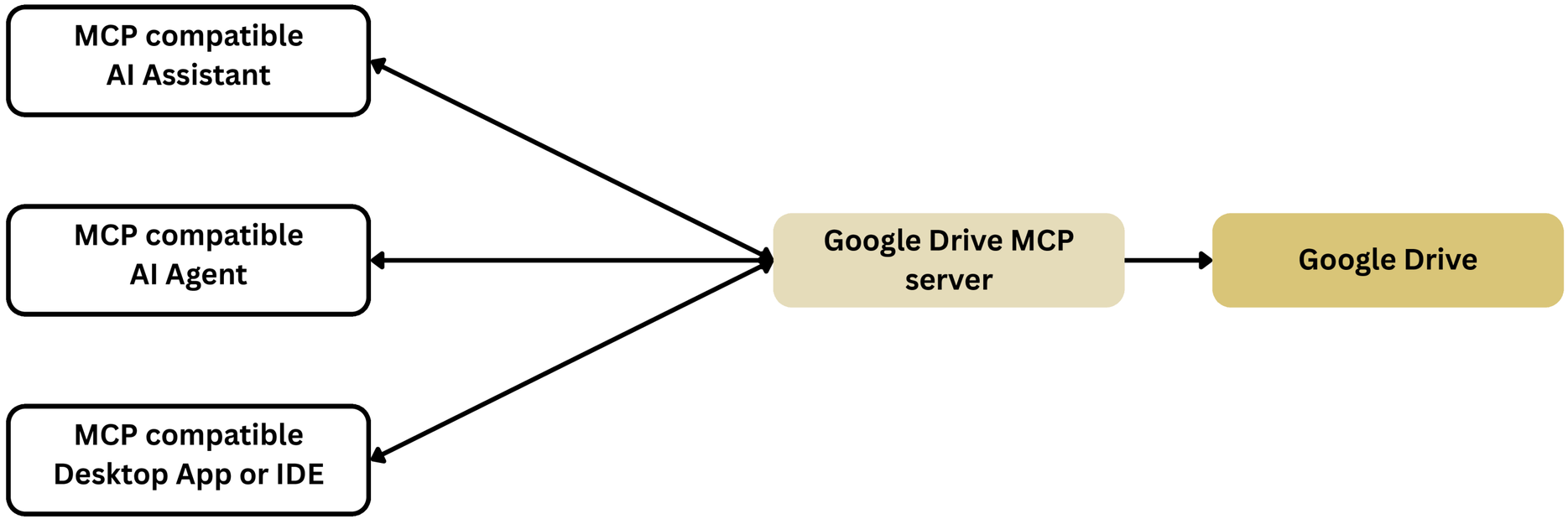
Why MCP

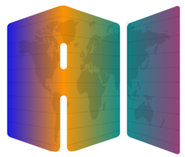
With MCP: Standardized AI Development



Why MCP

With MCP: MCP servers are reusable by various AI applications





Why MCP

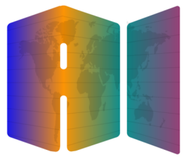
With MCP: Standardized AI Development

- **For AI application developers**
  Connect your app to any MCP server with 0 additional work

- **For tool or API developers**
  Build an MCP server once which can be adopted everywhere

- **For AI application users**
  AI applications have extensive capabilities

- **For enterprises**
  Clear separation of concerns between AI product teams



Why MCP

The MCP Ecosystem is Growing Fast

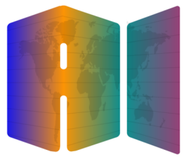
AI Applications, from IDEs to Agents

Server Builders (3000+ and counting)

Enterprises, for internal and external use cases

Open Source Contributors

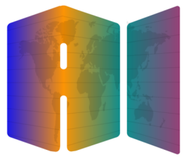
And a lot more...



Why MCP

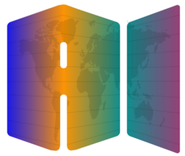
Common Questions

Who authors the MCP Server?	Anyone! Often the service provider itself will make their own MCP implementation. You can make an MCP server to wrap up access to some service.
How is using an MCP Server different from just calling a service's API directly?	MCP servers provide tool schemas + functions. If you want to directly call an API, you'll be authoring those on your own
Sounds like MCP Servers and tool use are the same thing	MCP Servers provide tool schemas + functions already defined for you



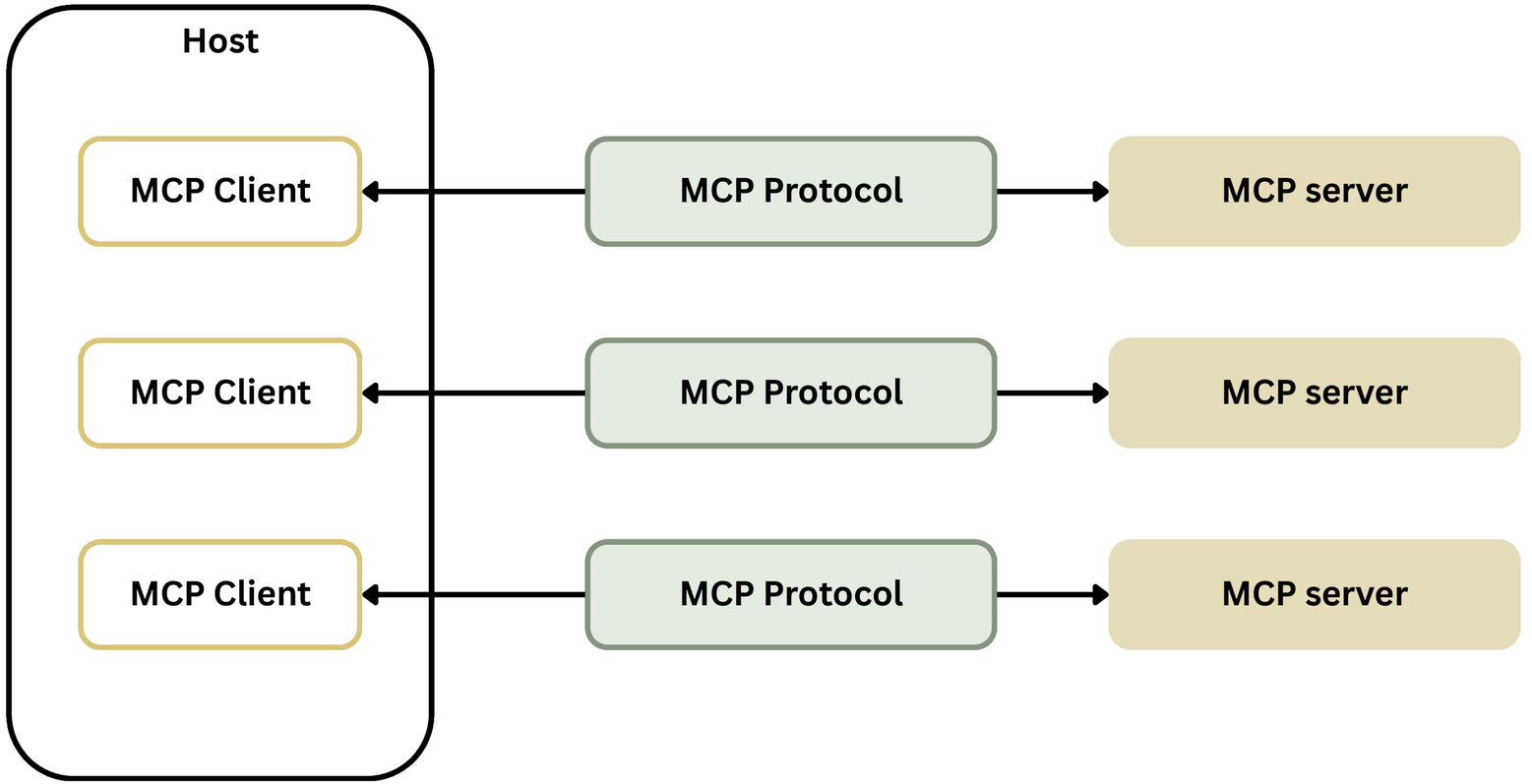
Model Context Protocol

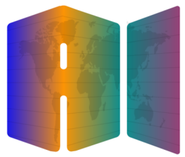
MCP Architecture



MCP Architecture

Client-Server Architecture





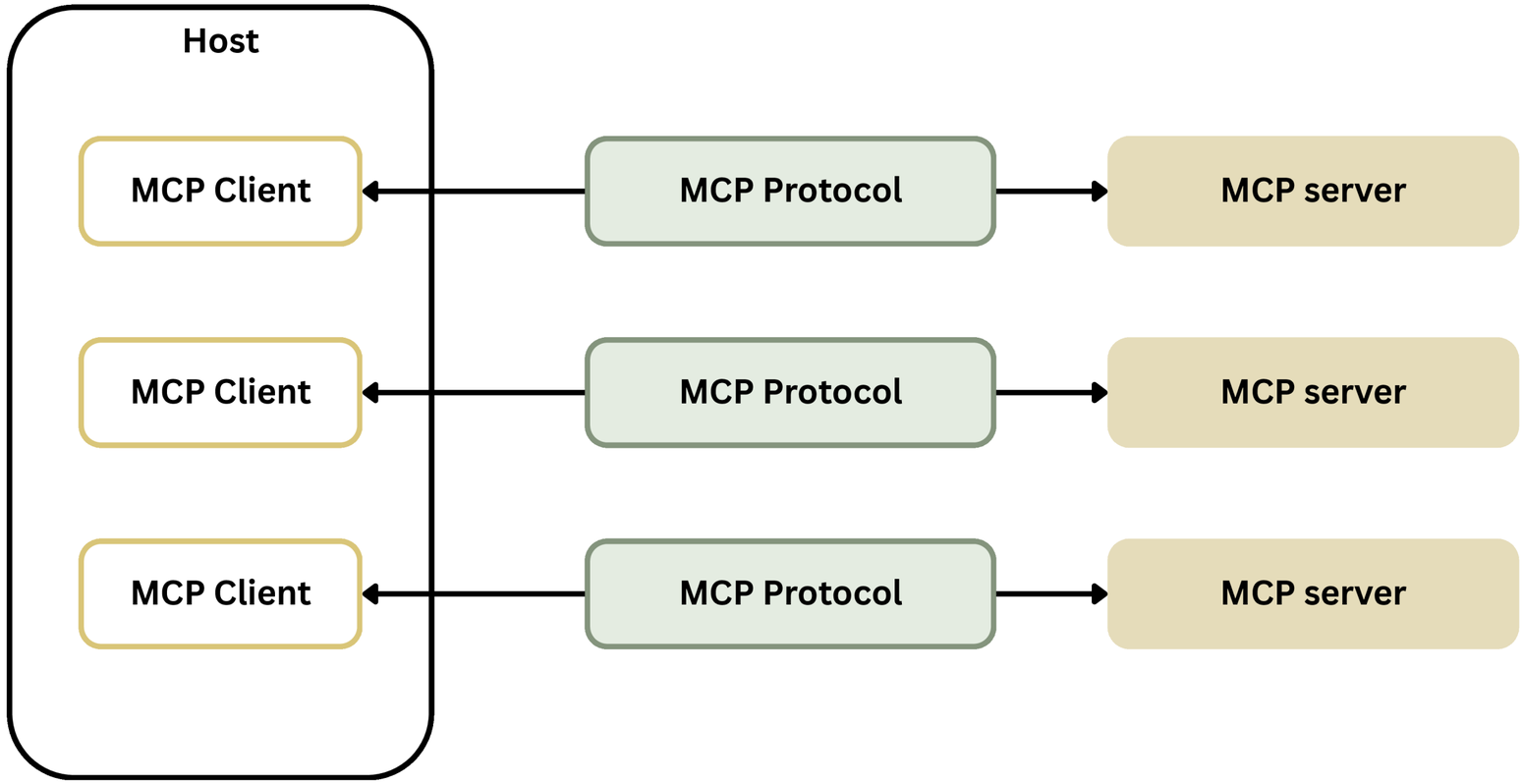
MCP Architecture

Client-Server Architecture

Hosts are LLM applications that want to access data through MCP (ex: Claude Desktops, IDEs, AI agents).

MCP Servers are lightweight programs that each expose specific capabilities through MCP.

MCP Clients maintain 1:1 connections with servers, inside the host application



MCP Architecture

How does it work?

MCP Client

Invokes **Tools**

Queries for **Resources**

Interpolates **Prompts**

MCP server

Exposes **Tools**

Exposes **Resources**

Exposes **Prompt Templates**

Tools

Functions and tools that can be invoked by the client

- Retrieve / search
- Send a message
- Update DB records

Resources

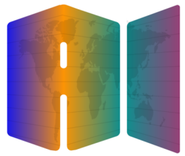
Read-only data or exposed by the server

- Files
- Database Records
- API Responses

Prompt Templates

Pre-defined templates for AI interactions

- Document Q&A
- Transcript Summary
- Output as JSON



MCP Architecture

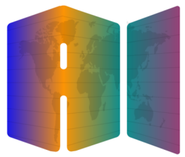
Defining a Tool

- MCP provides SDK’s for building servers and clients in a variety of languages
- We will use the Python MCP SDK
- The Python SDK makes it very easy to declare tools

MCP server

```python
@mcp.tool()
def add(a: int, b: int) → int:
    “””
    Args:
        a: First number to add
        b: Second number to add

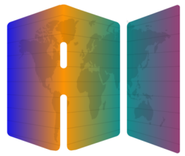
    Returns:
        The sum of the two numbers
    “””
    return a + b
```



MCP Architecture

Resources

- Allow the MCP Server to expose data to the client
- Similar to GET request handlers in a typical HTTP server
- Can return any type of data - strings, JSON, binary, etc.
  - You set the `mime_type` to give the client a hint as to what data you are returning
- Two types: direct and templated



MCP server

Direct

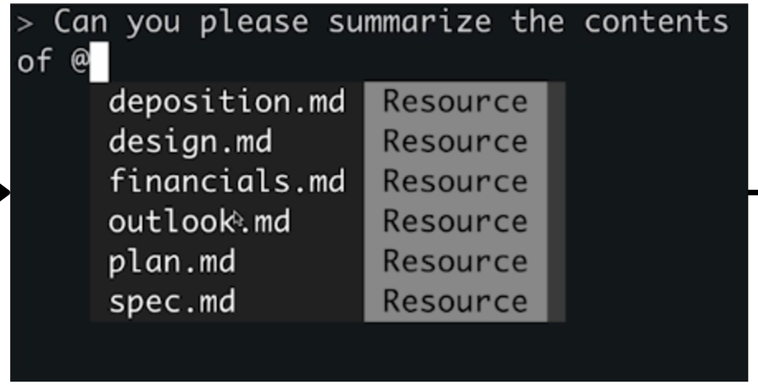
```
@mcp.resource(
    “docs://documents”,
    mime_type=“application/json”
)
def list_docs():
    # Return a list of document names
```

Templated

```
@mcp.resource(
    “docs://documents/{doc_id}”,
    mime_type=“text/plain”
)
def fetch_doc(doc_id: str):
    # Return the contents of a doc
```

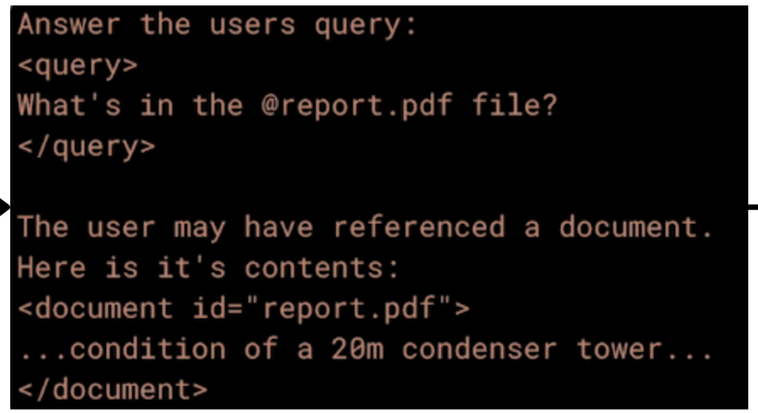
MCP Architecture

Direct resources

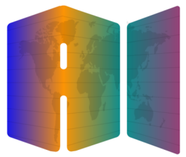


...You need the MCP Server to give a *list* of documents

Templated resources



...You need the MCP Server to give the contents of a *single* document



MCP Architecture

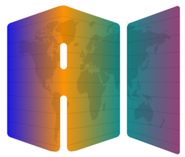
Prompts

- Defines a set of User and Assistant messages that can be used by the client
- These prompts should be high quality, and well-tested.

MCP server

Prompt

```
@mcp.prompt(
    name= “format”,
    description= “Rewrites the
contents of a document in Markdown
format”,
)
def format_document(
    doc_id: str,
) → list[base.Message]:
    # Return a list of messages
```



MCP Architecture

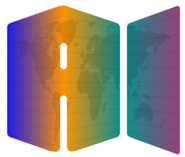
If you left this process up to a user, here's what they might write:

```
Convert report.pdf to markdown
```

Yes, it'd work, but the user might get a better result with some strong prompt engineering

They have more luck if they use a thoroughly evaluated prompt instead!

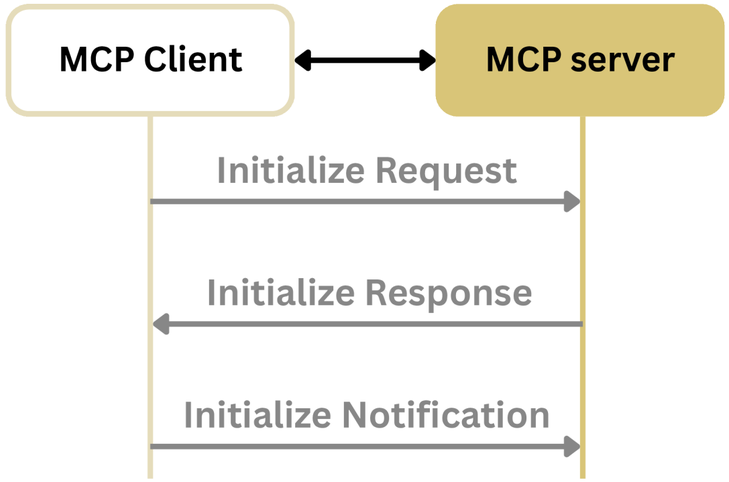
```
You are a document conversion specialist tasked with rewriting documents in Markdown
format. Your goal is to take the content of a given document and convert it into
well-structured Markdown, preserving the original meaning and enhancing readability.
Here is the identifier of the document you need to convert:
<document_id>
{{doc_id}}
</document_id>
Instructions:
1. Retrieve the content of the document associated with the given document_id.
2. Analyze the structure and content of the document.
3. Convert the document to Markdown format, following these guidelines:
- Use appropriate header levels (# for main titles, ## for subtitles, etc.)
- Properly format lists (both ordered and unordered)
- Use emphasis (*italic* or **bold**) where appropriate
- Add links and images using Markdown syntax if present in the original document
- Preserve any special formatting or structure that's important to the document's
meaning
Before providing the final Markdown output, in <document_analysis> tags:
- Identify the main sections and subsections of the document
- Count the number of sections and subsections to ensure proper nesting of headers
This will help ensure a thorough and well-organized conversion.
After your analysis, present the converted document in Markdown format. Use ``` markers
to denote the beginning and end of the Markdown content.
Example output structure:
<document_analysis>
[Your analysis of the document structure and conversion plan]
</document_analysis>
```markdown
# Document Title
## Section 1
Content of section 1...
## Section 2
Content of section 2...
- List item 1
- List item 2
[Link text](https://example.com)
![Image description](image-url.jpg)
```
Please proceed with your analysis and conversion of the document.
```



MCP Architecture

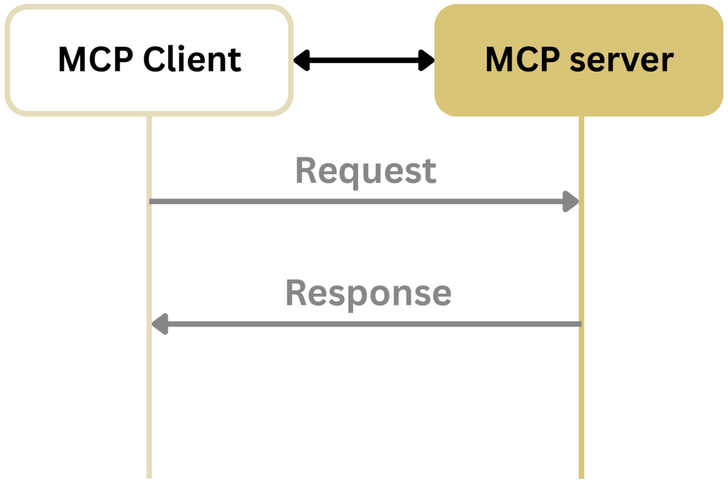
Communication Lifecycle

1. Initialization



Connection is established

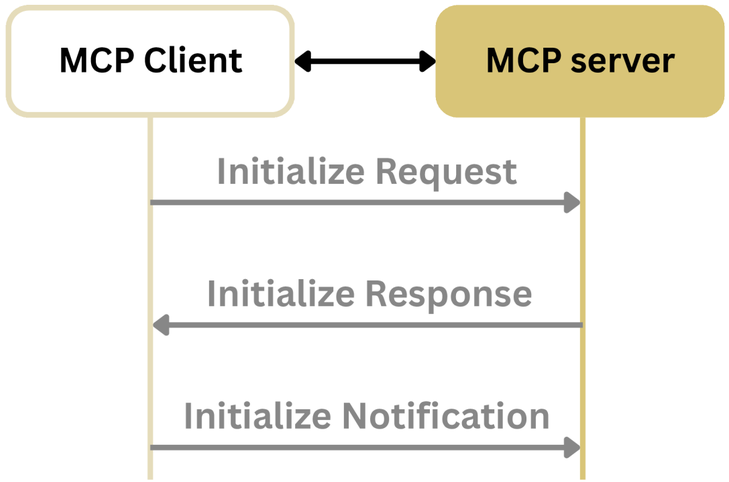
2. Message Exchange



MCP Architecture

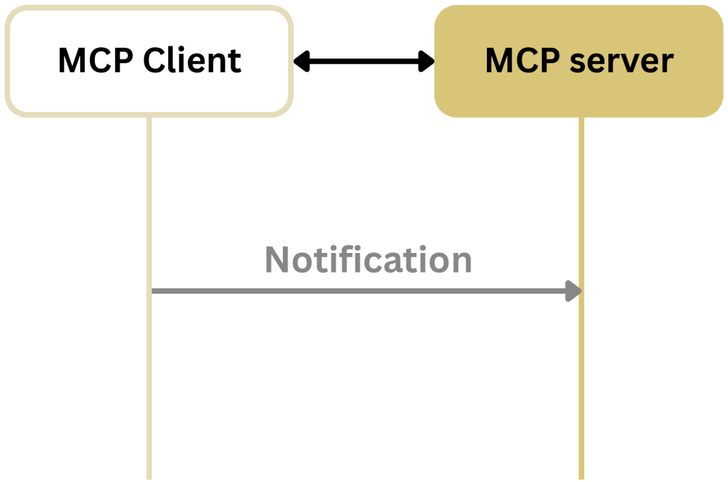
Communication Lifecycle

1. Initialization



Connection is established

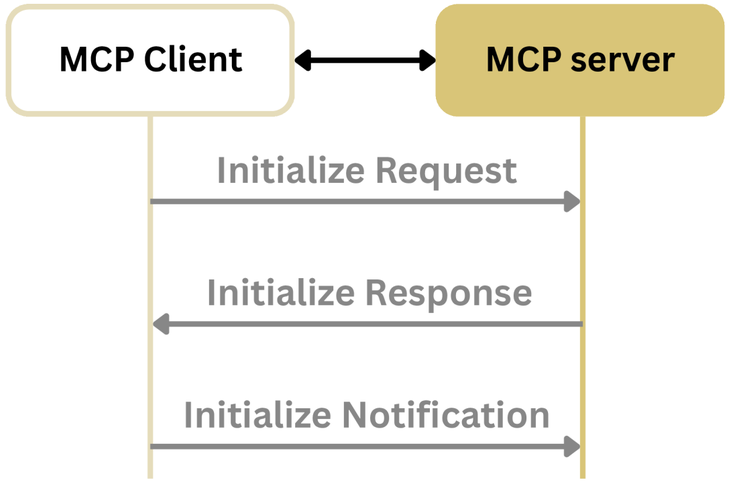
2. Message Exchange



MCP Architecture

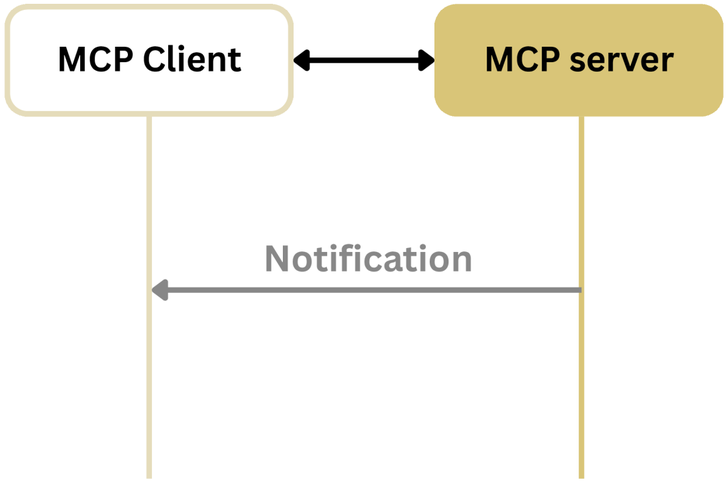
Communication Lifecycle

1. Initialization



Connection is established

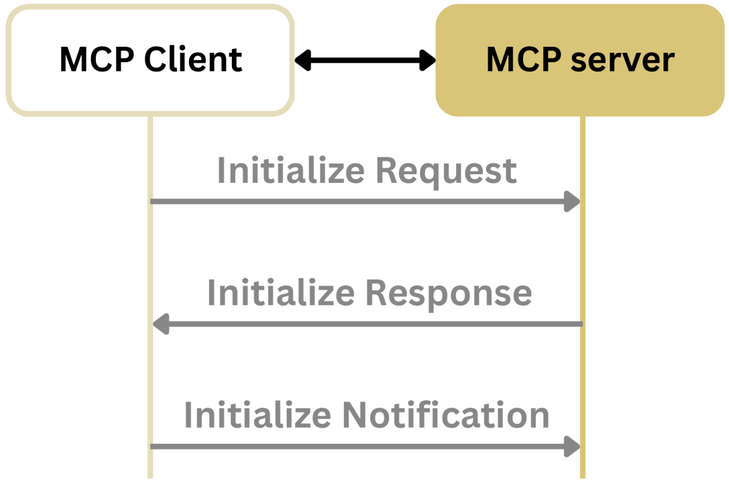
2. Message Exchange



MCP Architecture

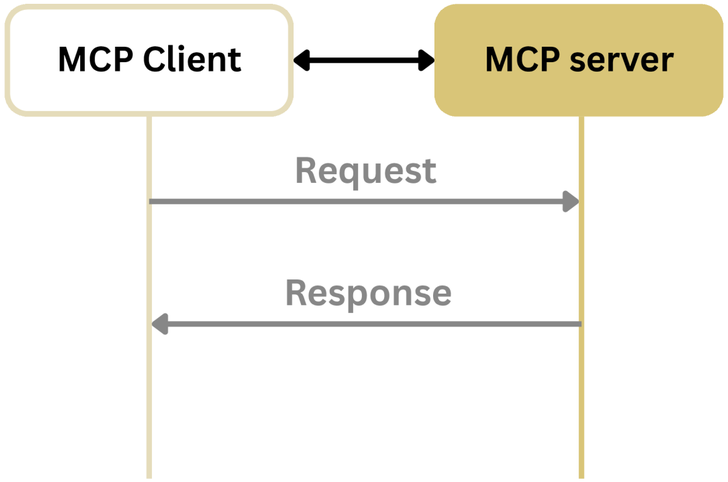
Communication Lifecycle

1. Initialization



Connection is established

2. Message Exchange



3. Termination

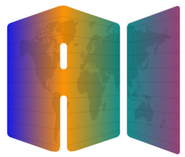
MCP Architecture

MCP Transports

A transport handles the underlying mechanics of how messages are sent and received between the client and server.

1. For servers running locally: **stdio** (standard input output)

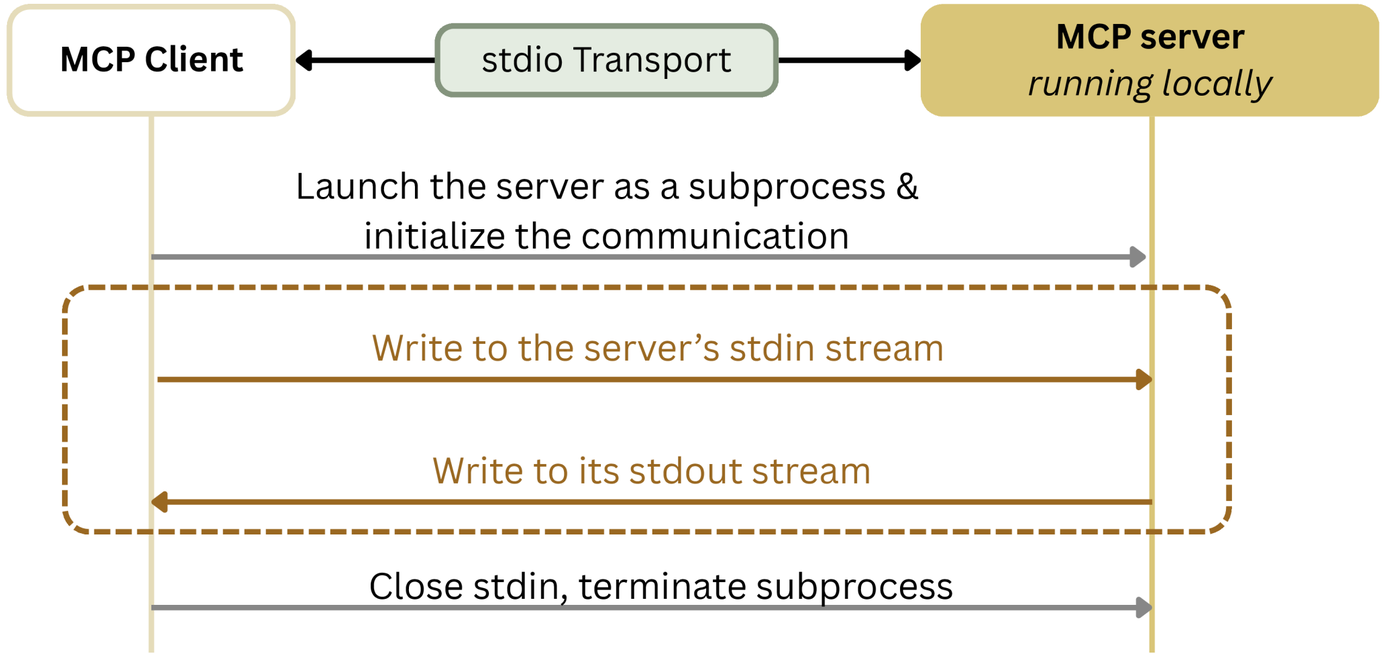
2. For remote servers:
    a. **HTTP+SSE (Server Sent Events)**
    b. **Streamable HTTP**



MCP Architecture

Standard IO (stdio) Transport

When running servers locally, stdio is most commonly used

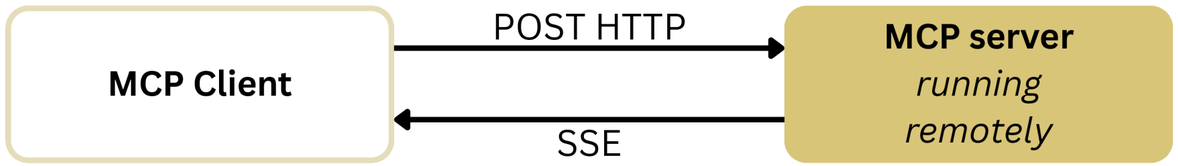


Message Exchange

MCP Architecture

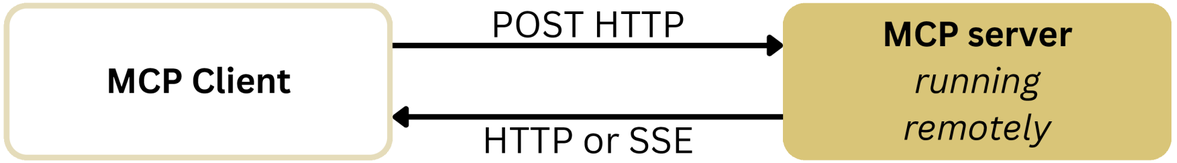
Transports for Remote Servers

HTTP + SSE



Stateful Connection

Streamable HTTP



Allow for Stateless or Stateful Connection

MCP Architecture

Streamable HTTP Transport

For remote MCP servers the Streamable HTTP transport is used. This supports stateless connections as well as stateful connections with the ability to opt into Server Sent Events